

CL219 Database Systems Lab
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Department of Computer Science
FAST-NU, Lahore, Pakistan

Project Title: Flex2
Student Learning Management System

Section: BCS-4H

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1. Introduction

Flex2 is a web-based Learning Management System (LMS) designed as a complete remake of the existing FAST-NU student portal. It provides a single, unified platform for Students, Teachers, and Administrators to manage courses, assignments, grades, attendance, and communication. The system is backed by a fully normalized relational database implemented in Microsoft SQL Server.

2. Functionalities

The following section details all the functionalities proposed for this project.

2.1. User Registration and Login

New users register by providing their name, email, password, phone number, and role (Student, Teacher, or Admin). The system validates that the email is unique before creating the account in the Users table. All users log in with their email and password; role-based access then determines which pages and actions they can see.

2.2. Profile Management

Any logged-in user can view and update their own profile: name, phone number, date of birth, and profile picture path. Students and teachers cannot change their role; only an Admin can promote or deactivate accounts via the IsActive flag.

2.3. Course Catalogue

A read-only catalogue lists all active courses from the Courses table, grouped by department. Each entry shows the course code, name, credit hours, description, and prerequisites. Students browse this catalogue to decide which offerings to enroll in.

2.4. Course Enrollment

Students select an active CourseOffering (semester, year, section) and enroll. The system checks that the offering has not reached its MaxCapacity and that the student is not already enrolled (enforced by the UQ_Enrollment constraint). Students can also drop an enrollment, which updates the Status field to 'Dropped'.

2.5. Assignment Management

Teachers create assignments for their course offerings, specifying title, type (Homework, Quiz, Exam, Project, or Lab), total marks, due date, and an optional file attachment path. Students enrolled in that offering can view all posted assignments from their dashboard.

2.6. Submissions and Grading

Students submit assignments by uploading a file or entering text content. Each student may submit once per assignment (UQ_Submission). Teachers review submissions, enter marks and written feedback, and update the status to 'Graded'. Late submissions are automatically flagged based on the due date.

2.7. Attendance Tracking

Teachers mark attendance per session for each enrolled student, recording status as Present, Absent, Late, or Excused. The unique constraint on (OfferingID, StudentID, AttendanceDate) prevents duplicate entries. Students can view their own attendance summary per course.

2.8. Grade Viewing

Once a teacher enters a final grade against an Enrollment record, the student can view it from their portal. Grades follow the standard letter scale stored in the Grade column (A+, A, A-, B+, B, B-, C+, C, C-, D, F, W, I). Admins can view all grades across offerings for reporting purposes.

2.9. Announcements

Teachers post course-level announcements with a title, body, and priority (Low, Normal, High, Urgent). Only students enrolled in the corresponding offering see these announcements. This covers notifications such as deadline extensions, lab room changes, and exam reminders.

2.10. Course Materials

Teachers upload and organise learning resources per offering. Each material has a title, type (Lecture, Notes, Video, Document, Link, or Other), and a file path or URL. Students see a listing of all materials for their enrolled courses and can follow the link or download the file.

2.11. Direct Messaging

Any user can send a private message to any other user on the platform. Messages have a subject and body and are stored in the Messages table with read/unread status and timestamps. Users see an inbox of received messages and can reply, creating a new message record.

3. Database Design Overview

The database Flex_2 is implemented in Microsoft SQL Server and contains the following eleven tables:

Table	Purpose
Users	All platform users with role (Student, Teacher, Admin).
Departments	Academic departments; each can have a head teacher.
Courses	Course catalogue linked to departments with credit hours.
CourseOfferings	Semester/section instances of a course assigned to a teacher.
Enrollments	Student–offering links, tracking grade and enrollment status.
Assignments	Assignments created per offering with type, marks, and due date.
Submissions	Student submissions per assignment with marks and feedback.
Attendance	Daily per-student attendance records for each offering.
Announcements	Course-level notices posted by teachers with priority levels.
CourseMaterials	Uploaded resources (slides, videos, documents) per offering.
Messages	Private messages between any two distinct platform users.

4. Tools and Technologies

Component	Technology
Database	Microsoft SQL Server 2022 with T-SQL.
Backend	Python (Flask) — lightweight REST API.
Frontend	HTML5, CSS3, JavaScript — simple responsive UI.
Version Control	Git / GitHub for team collaboration.
IDE / Tools	SQL Server Management Studio (SSMS), VS Code.

5. Conclusion

Flex2 is a comprehensive LMS that covers the core academic workflows at FAST-NU: user management, course enrollment, assignment submission and grading, attendance tracking, announcements, course materials sharing, and direct messaging. The platform brings together students, teachers, and administrators under one system, eliminating the need for separate tools and providing a centralized record of all academic activity across departments and semesters.

